

④ Select data points to send to next stump

⑤ Iterate process selecting random value between 0 and 1

<u>Salary</u>	<u>Credit</u>	<u>Approval</u>	<u>Random</u>
> 50K	N	yes	0.50
<= 50K	G	yes	0.10
> 50K	N	yes	0.60
> 50K	N	yes	0.75
<= 50K	G	yes	0.24
> 50K	B	No	0.32
> 50K	N	yes	0.87

⑥ These records will be sent to next DT stump

Salary	Credit	Approval	Sample weight
> 50K	Y	yes	1/6
<= 50K	Y	yes	1/6
> 50K	N	yes	1/6
> 50K	N	yes	1/6
<= 50K	Y	yes	1/6
> 50K	N	no	1/6
> 50K	N	yes	1/6

TE (Total Error)

Performance stump = 0.65

$$F_1 = \alpha_1 (M_1) + \alpha_2 (M_2)$$

$$\alpha_1 = 0.896$$

$$\alpha_2 = 0.65$$

Note: If calculate you will get α_2 .

$$= \frac{1}{2} \ln(n) \approx 0.805$$